

```
In [1]: import pandas as pd                #(for dataframes)
import numpy as np                      #(for numerical computation)
import matplotlib.pyplot as plt        #(for plotting)
import seaborn as sns                  #(for data visuvalization)
import plotly.express as px            #(for interactive scatter plot)
import plotly.io as pio                #for configure how plots are displayed
                                         #(e.g., in notebooks, browsers, or exported))

pio.renderers.default = "svg"          #(it will render as an SVG image by default)
```

```
In [2]: #!pip install plotly
```

```
In [3]: raw_df=pd.read_csv('Zomato_Mumbai_Dataset.csv',delimiter='|')
#(delimiter - split each row into columns wherever you see a pipe symbol)
```

```
In [4]: raw_df.head() #(first 5 rows)
```

```
Out[4]:
```

	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RA
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Biry...	Mumbai	First International Financial Centre--Bandra ...	https://www.zomato.com/mumbai/hitchki-bandra-k...	1	Casual Dining	12noon to 130am(Mon-Sun)	
1	Baba Falooda	400	Desserts,Ice Cream,Beverages	Mumbai	Mahim	https://www.zomato.com/mumbai/baba-falooda-mah...	1	Dessert Parlor	2pm to 1am(Mon-Sun)	
2	Chin Chin Chu	1800	Asian,Chinese	Mumbai	Juhu	https://www.zomato.com/mumbai/chin-chin-chu-ju...	1	Casual Dining	12noon to 1am(Mon-Sun)	
3	Butterfly High	1000	Modern Indian	Mumbai	Bandra Kurla Complex	https://www.zomato.com/mumbai/butterfly-high-b...	1	Bar	12noon to 130am(Mon-Sun)	
4	BKC DIVE	1200	North Indian,Chinese,Continental	Mumbai	Bandra Kurla Complex	https://www.zomato.com/mumbai/bkc-dive-bandra-...	1	Bar	1130am to 1am(Mon-Sun)	

3.Getting basic information about the Dataset

```
In [5]: raw_df.shape #(here we can know about total column and row in this case 15081 rows and 12 columns)
```

```
Out[5]: (15081, 12)
```

```
In [6]: raw_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15081 entries, 0 to 15080
Data columns (total 12 columns):
NAME                15081 non-null object
PRICE               15080 non-null object
CUSINE_CATEGORY     15079 non-null object
CITY                15080 non-null object
REGION              15080 non-null object
URL                 15080 non-null object
PAGE NO             15080 non-null object
CUSINE TYPE         15080 non-null object
TIMING              15015 non-null object
RATING_TYPE         15080 non-null object
RATING              15080 non-null object
VOTES               15080 non-null object
dtypes: object(12)
memory usage: 1.4+ MB
```

```
In [7]: raw_df.describe() #(in this table we can find how many duplicates entries are there among total count)
```

```
Out[7]:
```

	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
count	15081	15080	15079	15080	15080	15080	15080	15080	15015	15080	15080	15080
unique	12720	67	3183	2	241	13823	944	23	2551	32	35	1124
top	NAME	400	CUSINE_CATEGORY	Mumbai	REGION	URL	PAGE NO	Quick Bites	11am to 11pm(Mon-Sun)	Average	-	-
freq	942	2042	942	14138	942	942	942	5262	1192	5112	2360	2360

4. cleaning the dataset

removing duplicate rows of dataset

check for duplicates

```
In [8]: wrong_data=raw_df['PAGE NO']== 'PAGE NO'
```

```
In [9]: wrong_data
```

```
Out[9]: 0      False
1      False
2      False
3      False
4      False
...
15076   False
15077   False
15078   False
15079   False
15080   False
Name: PAGE NO, Length: 15081, dtype: bool
```

```
In [10]: raw_df[wrong_data]
```

```
Out[10]:
```

	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
15	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
31	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
47	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
63	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
79	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
...	...	...	...	...	...	...	...	...	...	...	...	...
15000	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
15016	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
15032	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
15048	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
15064	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES

942 rows × 12 columns

```
In [11]: #fliping the true values of the wrong data and then sorting the correct databack in the raw_df dataframe
#this permantly remove the wrong data from the original dataframe
# why fliping wrong dataset is nessasary?
## here as a bool wrong (True) and valid (False)
## So we have to flip right?
## thats why we use ~ (tilde operator) this is for Remove all duplicate only keep unique values
```

```
In [12]: raw_df=raw_df[~wrong_data]
```

```
In [13]: raw_df
```

Out[13]:

	NAME	PRICE	CUSINE_CATEGORY	CITY	REGION	URL	PAGE NO	CUSINE TYPE	
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	Mumbai	First International Financial Centre--Bandra ...	https://www.zomato.com/mumbai/hitchki-bandra-k...	1	Casual Dining	12
1	Baba Falooda	400	Desserts,Ice Cream,Beverages	Mumbai	Mahim	https://www.zomato.com/mumbai/baba-falooda-mah...	1	Dessert Parlor	
2	Chin Chin Chu	1800	Asian,Chinese	Mumbai	Juhu	https://www.zomato.com/mumbai/chin-chin-chu-ju...	1	Casual Dining	
3	Butterfly High	1000	Modern Indian	Mumbai	Bandra Kurla Complex	https://www.zomato.com/mumbai/butterfly-high-b...	1	Bar	12
4	BKC DIVE	1200	North Indian,Chinese,Continental	Mumbai	Bandra Kurla Complex	https://www.zomato.com/mumbai/bkc-dive-bandra-...	1	Bar	
...	...	...	...	...	...	...	...	...	
15076	Hari Om Snack Bar	350	Fast Food,South Indian,Chinese	Mumbai	Kandivali West	https://www.zomato.com/mumbai/hari-om-snack-ba...	99	Quick Bites	
15077	PitaBurg	400	Fast Food,Lebanese	Mumbai	Lower Parel	https://www.zomato.com/mumbai/pitaburg-lower-p...	99	none	11pm(Mon
15078	Uncha Otlawala	300	Desserts,Ice Cream	Mumbai	Kandivali West	https://www.zomato.com/mumbai/uncha-otlawala-1...	99	Dessert Parlor	9
15079	Mandarin Panda	400	Desserts,Chinese,Thai	Mumbai	Malad West	https://www.zomato.com/mumbai/mandarin-panda-m...	99	none	12noon to
15080		NaN	NaN	NaN	NaN	NaN	NaN	NaN	

14139 rows × 12 columns

In [14]:

```
## Dropping columns which are not required for further analysis
raw_df.drop(['URL','PAGE NO','CITY'],axis=1,inplace=True)
```

In [15]:

```
raw_df.head()
```

Out[15]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	First International Financial Centre--Bandra ...	Casual Dining	12noon to 130am(Mon-Sun)	Excellent	4.9	3529
1	Baba Falooda	400	Desserts,Ice Cream,Beverages	Mahim	Dessert Parlor	2pm to 1am(Mon-Sun)	Very Good	4.4	1723
2	Chin Chin Chu	1800	Asian,Chinese	Juhu	Casual Dining	12noon to 1am(Mon-Sun)	Very Good	4.2	337
3	Butterfly High	1000	Modern Indian	Bandra Kurla Complex	Bar	12noon to 130am(Mon-Sun)	Very Good	4.3	1200
4	BKC DIVE	1200	North Indian,Chinese,Continental	Bandra Kurla Complex	Bar	1130am to 1am(Mon-Sun)	Velmi dobré	4.4	5995

In [16]:

```
## Removing the null records
raw_df.isnull().sum()
```

Out[16]:

NAME	0
PRICE	1
CUSINE_CATEGORY	2
REGION	1
CUSINE TYPE	1
TIMING	66
RATING_TYPE	1
RATING	1
VOTES	1
dtype:	int64

In [17]:

```
## checking for null rows
raw_df[raw_df['PRICE'].isnull()]
```

Out[17]:

NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES
15080	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

In [18]:

```
## Drop the above NaN row from the raw_df
raw_df=raw_df.drop(labels=15080,axis=0)
```

In [19]:

```
## Replace the null(missing values) values with NA (not available)
```

```
raw_df.fillna('NA',inplace=True)
```

```
In [20]: ## check for null values
raw_df.isnull().sum()
```

```
Out[20]: NAME                0
PRICE                0
CUSINE_CATEGORY      0
REGION              0
CUSINE_TYPE          0
TIMING              0
RATING_TYPE          0
RATING              0
VOTES               0
dtype: int64
```

```
In [21]: ## Now our dataset has no null values but we have to check for datatype
raw_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 14138 entries, 0 to 15079
Data columns (total 9 columns):
NAME                14138 non-null object
PRICE              14138 non-null object
CUSINE_CATEGORY     14138 non-null object
REGION            14138 non-null object
CUSINE_TYPE        14138 non-null object
TIMING            14138 non-null object
RATING_TYPE        14138 non-null object
RATING            14138 non-null object
VOTES             14138 non-null object
dtypes: object(9)
memory usage: 1.1+ MB
```

```
In [22]: ## Here all datatypes are object but we have numerical columns too
## so we should convert the datatype of numerical columns to numeric datatype.
## checking for text values in the column before converting it to numeric datatype.
raw_df['RATING'].value_counts()
```

```
Out[22]: -                2360
3.5                1094
3.4                1036
3.6                 960
NEW                 953
3.3                 926
3.7                 917
3.2                 801
3.8                 782
3.1                 734
3.0                 622
3.9                 596
2.9                 409
4.0                 408
2.8                 309
4.1                 298
4.2                 199
2.7                 170
4.3                 148
4.4                  99
2.6                  77
Opening             57
4.5                  46
2.5                  39
4.6                  32
2.4                  26
4.7                  13
2.3                   10
2.1                    5
2.2                    4
4.8                    4
4.9                    2
2.0                    1
1.8                    1
Name: RATING, dtype: int64
```

```
In [23]: ## Replace the string (-,New,Opening) with "0"
## why rating should be in the numerical column so we have to change all string into numerical
raw_df['RATING'].replace(to_replace=['-','NEW','Opening'],value='0',inplace=True)
```

```
In [24]: ## checking for Price column
raw_df['PRICE'].value_counts()
```

```
Out[24]: 400      2042
          500      1954
          300      1936
          200      1177
          600      1062
          ...
          80         1
          1850        1
          3200        1
          248         1
          360         1
Name: PRICE, Length: 66, dtype: int64
```

```
In [25]: raw_df['PRICE'].dtype
```

```
Out[25]: dtype('O')
```

Replace strings in Price column to convert it into numerical

```
import pandas as pd
pd.set_option('display.max_seq_items', None)
raw_df['PRICE'].unique()
```

Output values numbers are in string so that it shows object.

```
array(['1200', '400', '1800', '1000', '800', '1300', '1500', '600', '1400', '1100', '2000', '350', '900', '700', '500', '1600', '150', '300', '550', '450', '650', '750', '250', '200', '850', '100', '2500', '1450', '180', '950', '1700', '1900', '2600', '2400', '3000', '1250', '3200', '505', '2200', '1050', '1650', '1350', '50', '920', '1150', '1550', '3500', '398', '5000', '120', '220', '160', '340', '5', '480', '2100', '580', '1850', '249', '80', '2300', '0', '360', '3502', '248', '2700'], dtype=object)
```

In this case we just convert the dtype from object to int thats enough.

```
In [26]: ## checking for votes column
raw_df['VOTES'].value_counts()
```

```
Out[26]: -      2360
NEW      953
4        364
5        320
6        288
          ...
634         1
1653         1
2528         1
1117         1
573          1
Name: VOTES, Length: 1123, dtype: int64
```

```
In [27]: ## Replace strings in VOTES column to convert it into numerical
raw_df['VOTES'].replace(to_replace=['-', 'NEW', 'Opening'], value='0', inplace=True)
```

```
In [28]: ## Changing datatype of the numerical columns as int or float
raw_df['PRICE'] = raw_df['PRICE'].astype('int64')
raw_df['VOTES'] = raw_df['VOTES'].astype('int64')
raw_df['RATING'] = raw_df['RATING'].astype('float64')
```

```
In [29]: raw_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 14138 entries, 0 to 15079
Data columns (total 9 columns):
NAME      14138 non-null object
PRICE     14138 non-null int64
Cuisine_CATEGORY  14138 non-null object
REGION    14138 non-null object
Cuisine_Type  14138 non-null object
TIMING    14138 non-null object
RATING_Type  14138 non-null object
RATING    14138 non-null float64
VOTES     14138 non-null int64
dtypes: float64(1), int64(2), object(6)
memory usage: 1.1+ MB
```

```
In [30]: ## working with timing
raw_df['TIMING'].value_counts()
```

```
Out[30]: 11am to 11pm(Mon-Sun) 1192
11am to 12midnight(Mon-Sun) 632
12noon to 12midnight(Mon-Sun) 467
11am to 1130pm(Mon-Sun) 309
10am to 10pm(Mon-Sun) 267
...
Closed(Mon),9am to 12midnight(Tue,Wed,Fri,Sat... 1
12noon to 4pm,7pm to 12midnight(Mon-Thu),12noon... 1
11am to 4pm,6pm to 1230AM(Mon,Tue,Wed,Thu... 1
9am to 10pm(Mon-Fri),9am to 12midnight(Sat-Sun) 1
1pm to 430pm,7pm to 11pm(Mon-Sun) 1
Name: TIMING, Length: 2551, dtype: int64
```

```
In [31]: ## split the columns and storing it in tem_df temporary dataframe
tem_df=raw_df['TIMING'].str.split("(",n=1,expand=True)
```

```
In [32]: tem_df
```

Out[32]:

	0	1
0	12noon to 130am	Mon-Sun)
1	2pm to 1am	Mon-Sun)
2	12noon to 1am	Mon-Sun)
3	12noon to 130am	Mon-Sun)
4	1130am to 1am	Mon-Sun)
...	...	...
15075	8am to 11pm,12midnight to 115am	Mon-Sun)
15076	11am to 230am	Mon-Sun)
15077	11am to 11pm Mon,Tue,Wed,Thu,Sun),11am to ...	
15078	9am to 1230AM	Mon-Sun)
15079	12noon to 330pm,7pm to 1am	Mon-Sun)

14138 rows × 2 columns

```
In [33]: ## Assigning the columns back to the raw_df dataframe
raw_df['TIMING']=tem_df[0]
raw_df['DAYS_OPENS']=tem_df[1]
raw_df.head()
```

Out[33]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE_TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPENS
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	First International Financial Centre--Bandra ...	Casual Dining	12noon to 130am	Excellent	4.9	3529	Mon-Sun)
1	Baba Falooda	400	Desserts,Ice Cream,Beverages	Mahim	Dessert Parlor	2pm to 1am	Very Good	4.4	1723	Mon-Sun)
2	Chin Chin Chu	1800	Asian,Chinese	Juhu	Casual Dining	12noon to 1am	Very Good	4.2	337	Mon-Sun)
3	Butterfly High	1000	Modern Indian	Bandra Kurla Complex	Bar	12noon to 130am	Very Good	4.3	1200	Mon-Sun)
4	BKC DIVE	1200	North Indian,Chinese,Continental	Bandra Kurla Complex	Bar	1130am to 1am	Vefmi dobré	4.4	5995	Mon-Sun)

```
In [34]: ## Good now we are going to remove the brackets from days column
raw_df['DAYS_OPENS']=raw_df['DAYS_OPENS'].str.replace(')','',regex=True)
```

```
In [35]: raw_df.head()
```

Out[35]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE_TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPENS
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	First International Financial Centre--Bandra ...	Casual Dining	12noon to 130am	Excellent	4.9	3529	Mon-Sun
1	Baba Falooda	400	Desserts,Ice Cream,Beverages	Mahim	Dessert Parlor	2pm to 1am	Very Good	4.4	1723	Mon-Sun
2	Chin Chin Chu	1800	Asian,Chinese	Juhu	Casual Dining	12noon to 1am	Very Good	4.2	337	Mon-Sun
3	Butterfly High	1000	Modern Indian	Bandra Kurla Complex	Bar	12noon to 130am	Very Good	4.3	1200	Mon-Sun
4	BKC DIVE	1200	North Indian,Chinese,Continental	Bandra Kurla Complex	Bar	1130am to 1am	Vefmi dobré	4.4	5995	Mon-Sun

```
In [36]: ## Checking for null records in Days Open column
```

```
In [30]: ## checking for null records in days_open column
raw_df['DAYS_OPENS'].isnull().sum()
```

Out[36]: 160

```
In [37]: raw_df.isnull().sum()
```

Out[37]:

NAME	0
PRICE	0
CUSINE_CATEGORY	0
REGION	0
CUSINE TYPE	0
TIMING	0
RATING_TYPE	0
RATING	0
VOTES	0
DAYS_OPENS	160

dtype: int64

```
In [38]: ## we have to replace this 160 No entries values with NA
raw_df.fillna('NA',inplace=True)
```

```
In [39]: ## check for dtypes for all columns
raw_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 14138 entries, 0 to 15079
Data columns (total 10 columns):
NAME                14138 non-null object
PRICE               14138 non-null int64
CUSINE_CATEGORY     14138 non-null object
REGION              14138 non-null object
CUSINE TYPE         14138 non-null object
TIMING              14138 non-null object
RATING_TYPE         14138 non-null object
RATING              14138 non-null float64
VOTES               14138 non-null int64
DAYS_OPENS          14138 non-null object
dtypes: float64(1), int64(2), object(7)
memory usage: 1.2+ MB
```

```
In [40]: ## Removing the restaruant records whose Rating or votes is "0"
## Finding those restaruant whose has "0" as Rating or votes.
useless_data = (raw_df['RATING'] == 0.0) | (raw_df['VOTES'] == 0)
# | means "either condition can be true."
raw_df[useless_data]
```

Out[40]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPENS
32	Hotel Annapoorna Refreshments	400	Maharashtrian,Mughlai,Chinese	Ghansoli	Quick Bites	1030am to 1230AM	Not rated	0.0	0	Mon-Sun
34	Biryani 9	600	Biryani,North Indian	Near Andheri East Station	none	11am to 3am	None	0.0	0	Mon-Sun
36	D Fusion Flavours	350	Chinese	Goregaon East	none	12noon to 330pm,7pm to 3am	None	0.0	0	Mon-Sun
39	Nation Tadka	400	North Indian,South Indian,Chinese,Fast Food	Worli	none	12noon to 1230AM	Not rated	0.0	0	Mon-Sun
83	Link Way Restaurant	500	North Indian,Chinese	Jogeshwari	Quick Bites	12noon to 4pm,8pm to 1am	Not rated	0.0	0	Mon-Sun
...	...	...	...	...	...	...	...	...	...	...
14998	Foodies House	0	Chinese	Goregaon East	none	12noon to 4am	None	0.0	0	Mon-Sun
14999	Khansama	0	Biryani	Lower Parel	none	12noon to 3am	None	0.0	0	Mon-Sun
15010	Earth Cafe @ Waterfield	800	Cafe,Healthy Food,Italian,Pizza,Beverages	Linking Road--Bandra West	Café	10am to 10pm	None	0.0	0	Mon-Thu,10am to 11pm(Fri-Sun
15023	How About Some Cream	200	Beverages	Mumbai Central	Beverage Shop	12noon to 3am	None	0.0	0	Mon-Sun
15046	Food And Taste Theory	800	Continental,Italian	Phoenix Marketcity- Kurla	Casual Dining	9am to 12midnight	None	0.0	0	Mon-Fri,9am to 1230AM(Sat-Sun

3371 rows × 10 columns

```
In [41]: raw_df["RATING"].value_counts()
```

```
Out[41]: 0.0    3370
3.5    1094
3.4    1036
3.6     960
3.3     926
3.7     917
3.2     801
3.8     782
3.1     734
3.0     622
3.9     596
2.9     409
4.0     408
2.8     309
4.1     298
4.2     199
2.7     170
4.3     148
4.4      99
2.6      77
4.5      46
2.5      39
4.6      32
2.4      26
4.7      13
2.3      10
2.1       5
2.2       4
4.8       4
4.9       2
2.0       1
1.8       1
Name: RATING, dtype: int64
```

```
In [42]: ## Performing negation of the useless dataset and then sorting the correct data back in the raw_df
## this permanently remove the useless data from the original dataframe

raw_df=raw_df[~useless_data]
```

```
In [43]: raw_df
```

Out[43]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES	DA
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	First International Financial Centre-- Bandra ...	Casual Dining	12noon to 130am	Excellent	4.9	3529	
1	Baba Falooda	400	Desserts,Ice Cream,Beverages	Mahim	Dessert Parlor	2pm to 1am	Very Good	4.4	1723	
2	Chin Chin Chu	1800	Asian,Chinese	Juhu	Casual Dining	12noon to 1am	Very Good	4.2	337	
3	Butterfly High	1000	Modern Indian	Bandra Kurla Complex	Bar	12noon to 130am	Very Good	4.3	1200	
4	BKC DIVE	1200	North Indian,Chinese,Continental	Bandra Kurla Complex	Bar	1130am to 1am	Vel'mi dobré	4.4	5995	
...	...	...	...	...	...	...	...	...	...	
15075	Tirupati Balaji	500	Chinese,Fast Food,North Indian	Oshiwara-- Andheri West	Casual Dining	8am to 11pm,12midnight to 115am	Good	3.5	267	
15076	Hari Om Snack Bar	350	Fast Food,South Indian,Chinese	Kandivali West	Quick Bites	11am to 230am	Good	3.7	64	
15077	PitaBurg	400	Fast Food,Lebanese	Lower Parel	none	11am to 11pm	Average	3.4	99	Mon,Tue,Wed,Th
15078	Uncha Otlawala	300	Desserts,Ice Cream	Kandivali West	Dessert Parlor	9am to 1230AM	Good	3.5	29	
15079	Mandarin Panda	400	Desserts,Chinese,Thai	Malad West	none	12noon to 330pm,7pm to 1am	Good	3.7	121	

10767 rows × 10 columns

```
In [44]: ## Working on 'Rating type' column
## checking the unique values in the column
raw_df['RATING_TYPE'].value_counts()
```



```
Out[44]: Average      5111
Good      4330
Very Good  1137
Excellent   95
Poor       47
Vel'mi dobré    6
Skvělá volba    4
Dobrze         4
Bardzo dobrze  3
Bueno          2
İyi            2
Bom            2
Průměr         2
Priemer        2
Buono          2
Muito Bom      2
Promedio       2
Dobré          2
Ortalama       2
Velmi dobré    1
Excelente      1
Biasa          1
Sangat Baik    1
Media          1
Média          1
Muy Bueno      1
Çok iyi        1
Baik           1
Skvělé         1
Name: RATING_TYPE, dtype: int64
```

```
In [45]: ## In the above cell we got Unknown Name so we have to translate it to create visuvalization or analysis.
## Translating the texts into proper English text.
raw_df['RATING_TYPE'].replace(to_replace='Excelente' , value='Excellent', inplace=True)
raw_df['RATING_TYPE'].replace(to_replace=['Vel'mi dobré','Bardzo dobrze','Muy Bueno','Velmi dobré'] , value='Very Good', inplace=True)
raw_df['RATING_TYPE'].replace(to_replace=['Skvělá volba','Dobrze','Bueno','Buono','Dobré','Bom','Skvělé'] , value='Good', inplace=True)
raw_df['RATING_TYPE'].replace(to_replace=['Priemer','Média','Çok iyi'] , value='Average', inplace=True)
raw_df['RATING_TYPE'].replace(to_replace=['Průměr','Promedio','Ortalama','Muito Bom','İyi'] , value='Poor', inplace=True)
raw_df['RATING_TYPE'].replace(to_replace=['Baik','Biasa','Media','Sangat Baik'] , value='Very Poor', inplace=True)
```

```
In [46]: ## Check for correction (Translation)
raw_df['RATING_TYPE'].value_counts()
```

```
Out[46]: Average      5115
Good      4347
Very Good  1148
Excellent   96
Poor       57
Very Poor    4
Name: RATING_TYPE, dtype: int64
```

```
In [47]: ## Working on 'Regeion' column
raw_df['REGION'].value_counts()
```

```
Out[47]: Mira Road      405
Malad West      308
Chembur        277
Kharghar        268
Borivali West   264
...
Pallavi Avida-- Kalamboli    1
Atria Mall-- Worli          1
Fariyas Hotel-- Colaba      1
Hotel Heritage Supreme-- Vashi  1
Orchard Mall-- Goregaon East  1
Name: REGION, Length: 237, dtype: int64
```

'Hotel Jewel of Chembur-- Chembur' In this record we want only region name not hotel name

we should remove unwanted text in region column

How we remove unwanted text using regex (regular expression)

Regex decides what to keep,remove, or replace.

```
In [48]: ## Removing the irrelevant text from the region column
raw_df['REGION'] = raw_df['REGION'].str.replace('[a-zA-Z].+-- ','',regex=True)
#raw_df["REGION"]=raw_df["REGION"].str.replace('[a-zA-Z].+-- ', regex=True)
```

```
In [49]: ## Removing the west and east from the region column
raw_df['REGION']=raw_df["REGION"].str.replace('West|west|East|east','',regex=True)
```

```
In [50]: raw_df['REGION'].value_counts() # Irrelevant and unwanted texts are removed.

Out[50]: Thane                726
Mira Road                412
Andheri                  409
Malad                    378
Kandivali                377
...
Panvel                    2
Girgaon Chowpatty        1
CBD Belapur              1
Goregaon                 1
Dadar                    1
Name: REGION, Length: 104, dtype: int64

In [51]: ## Replacing small regions with know region names (know region already listed in region column)
raw_df['REGION'] = raw_df['REGION'].str.replace('4 Bungalows|7 Andheri|Azad Nagar|Near Andheri Station|Veera De
raw_df['REGION'] = raw_df['REGION'].str.replace('Bandra Kurla Complex', 'Bandra', regex=True)
raw_df['REGION'] = raw_df['REGION'].str.replace('CBD-Belapur', 'CBD Belapur', regex=True)
raw_df['REGION'] = raw_df['REGION'].str.replace('Girgaon Chowpatty', 'Chowpatty', regex=True)
raw_df['REGION'] = raw_df['REGION'].str.replace('Dadar Shivaji Park', 'Dadar', regex=True)
raw_df['REGION'] = raw_df['REGION'].str.replace('Flea Bazaar Café|Kamala Mills Compound', 'Lower Parel', regex=Tr
raw_df['REGION'] = raw_df['REGION'].str.replace('Runwal Green', 'Mulund', regex=True)
raw_df['REGION'] = raw_df['REGION'].str.replace('Mumbai CST Area', 'Mumbai Central', regex=True)
raw_df['REGION'] = raw_df['REGION'].str.replace('Kopar Khairane|Seawoods|Turbhe|Ulwe', 'Navi Mumbai', regex=True)
raw_df['REGION'] = raw_df['REGION'].str.replace('New Panvel|Old Panvel', 'Panvel', regex=True)
raw_df['REGION'] = raw_df['REGION'].str.replace('Kamothe', 'Sion', regex=True)
raw_df['REGION'] = raw_df['REGION'].str.replace('Ghodbunder Road|Majiwada', 'Thane', regex=True)
```

```
In [52]: raw_df.head()
```

Out[52]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPENS
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	Bandra	Casual Dining	12noon to 130am	Excellent	4.9	3529	Mon-Sun
1	Baba Falooda	400	Desserts,Ice Cream,Beverages	Mahim	Dessert Parlor	2pm to 1am	Very Good	4.4	1723	Mon-Sun
2	Chin Chin Chu	1800	Asian,Chinese	Juhu	Casual Dining	12noon to 1am	Very Good	4.2	337	Mon-Sun
3	Butterfly High	1000	Modern Indian	Bandra	Bar	12noon to 130am	Very Good	4.3	1200	Mon-Sun
4	BKC DIVE	1200	North Indian,Chinese,Continental	Bandra	Bar	1130am to 1am	Very Good	4.4	5995	Mon-Sun

```
In [53]: ## Removing duplicate records (Repeated names)
## Finding all the duplicate rows.
raw_df[raw_df.duplicated()] #It shows only duplicate same name,price everything so, we have to remove this dupl
```

Out[53]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPENS
4064	Sai Sannidhi Restaurant & Bar	1000	North Indian,Konkan	Dahisar	Casual Dining	11am to 12midnight	Good	3.7	99	Mon-Sun
4068	Konkan Katta	400	Seafood,Maharashtrian,Malwani	Mahakali	Quick Bites	11am to 330pm,630pm to 1130pm	Good	3.5	181	Mon-Sun
4082	Usmaniya Hotel	600	Mughlai	Fort	Casual Dining	1030am to 1130pm	Average	3.2	8	Mon-Sun
4083	Gina's Cakes	450	Bakery	Dombivali	none	11am to 11pm	Good	3.5	49	Mon-Sun
4084	Konkanastha Lunch Home	400	Seafood,Malwani	Chakala	Casual Dining	12noon to 3pm,730pm to 1030pm	Good	3.5	44	Mon-Sun
...	...	...	...	...	...	...	...	...	...	...
14200	Mezbaan Family Restaurant	350	Chinese,Mughlai	Mumbra	Dhaba	12noon to 1230AM	Average	2.8	97	Mon-Sun
14204	Jyoti Lunch Home	650	Chinese,North Indian,Seafood,Mughlai	Mulund	Casual Dining	11am to 1230AM	Good	3.5	49	Mon-Sun
14253	On Toes	900	Italian,North Indian,Chinese	Malad	Casual Dining	12noon to 3pm,7pm to 1230AM	Good	3.6	76	Mon-Sun
14761	Frosty Farm	400	Ice Cream,Desserts,Fast Food	Malad	Dessert Parlor	1pm to 1215AM	Good	3.6	120	Mon-Sun
14928	Shree Manu Sagar	300	North Indian,Chinese,Indian	Ghansoli	Quick Bites	1130am to 415pm,7pm to 1215AM	Average	3.3	51	Mon-Sun

220 rows × 10 columns

```
In [54]: ## dropping all the duplicate rows.  
raw_df=raw_df.drop_duplicates()
```

```
In [55]: raw_df.info()  
  
<class 'pandas.core.frame.DataFrame'>  
Int64Index: 10547 entries, 0 to 15079  
Data columns (total 10 columns):  
NAME                10547 non-null object  
PRICE               10547 non-null int64  
CUSINE_CATEGORY     10547 non-null object  
REGION              10547 non-null object  
CUSINE TYPE         10547 non-null object  
TIMING              10547 non-null object  
RATING_TYPE         10547 non-null object  
RATING              10547 non-null float64  
VOTES               10547 non-null int64  
DAYS_OPENS          10547 non-null object  
dtypes: float64(1), int64(2), object(7)  
memory usage: 906.4+ KB
```

```
In [56]: ## coping the cleaned dataset into a new dataframe  
zomato_df=raw_df.copy()  
zomato_df.head()
```

Out[56]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPENS
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Biry...	Bandra	Casual Dining	12noon to 130am	Excellent	4.9	3529	Mon-Sun
1	Baba Falooda	400	Desserts,Ice Cream,Beverages	Mahim	Dessert Parlor	2pm to 1am	Very Good	4.4	1723	Mon-Sun
2	Chin Chin Chu	1800	Asian,Chinese	Juhu	Casual Dining	12noon to 1am	Very Good	4.2	337	Mon-Sun
3	Butterfly High	1000	Modern Indian	Bandra	Bar	12noon to 130am	Very Good	4.3	1200	Mon-Sun
4	BKC DIVE	1200	North Indian,Chinese,Continental	Bandra	Bar	1130am to 1am	Very Good	4.4	5995	Mon-Sun

Performing exploratory data analysis

Q1) How many restaurants are in Mumbai for each type of cuisine? (Frequency-repeated)

Exploratory means _ both descriptive and inferential analysis

```
In [57]: #! pip install -U kaleido
```

```
In [58]: import pandas as pd
```

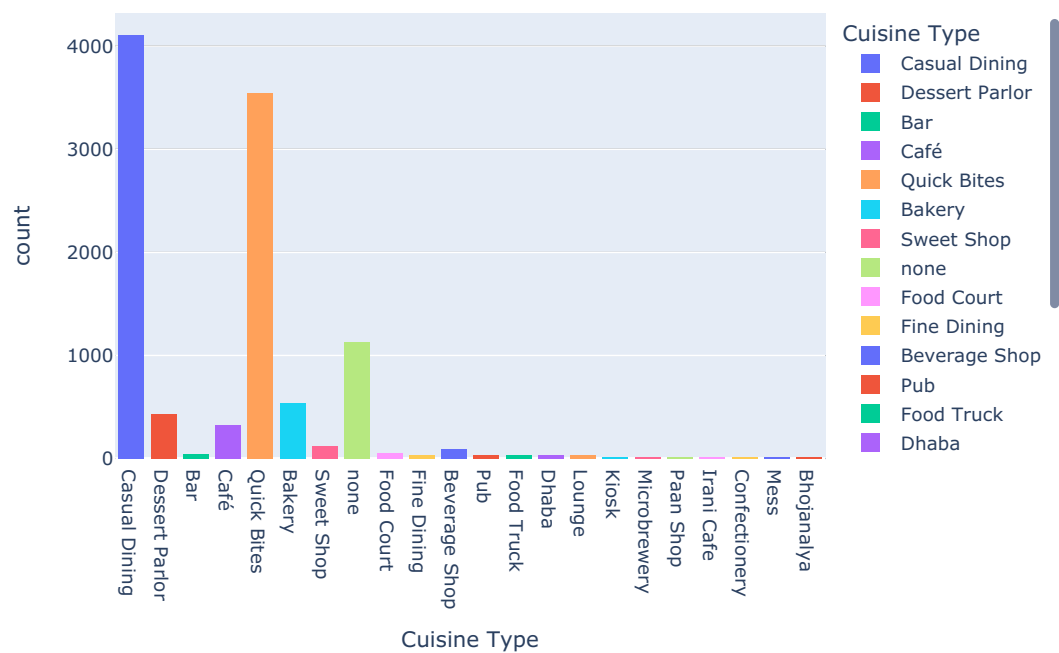
```
In [59]: import plotly.io as pio
pio.renderers.default = "notebook"
```

```
In [60]: fig = px.histogram(zomato_df, x='Cuisine Type', color='Cuisine Type',
                  title='No. of Restaurants by Cuisine Type',
                  labels={'Cuisine Type': 'Cuisine Type'})

fig.show()
```



No. of Restaurants by Cuisine Type



Q2, what are the percentage of restaurants by rating type in Mumbai? (Relative frequency)

If you want to tell the frequency in percentage we can use relative frequency.

```
In [61]: zomato_df['RATING_TYPE'].value_counts()
```

```
Out[61]:
```

Average	4983
Good	4263
Very Good	1145
Excellent	96
Poor	56
Very Poor	4

Name: RATING_TYPE, dtype: int64

```
In [62]: ## Reset index number for making new table.
zomato_df['RATING_TYPE'].value_counts().reset_index()
```

```
Out[62]:
```

	index	RATING_TYPE
0	Average	4983
1	Good	4263
2	Very Good	1145
3	Excellent	96
4	Poor	56
5	Very Poor	4

```
In [63]: ## Save this new table in related variable.
rating_type_df = zomato_df['RATING_TYPE'].value_counts().reset_index()
```

```
In [64]: #just rename index and rating_type for clarity.
rating_type_df.rename(columns={"index":"RATING_TYPE","RATING_TYPE":"Count_OF_RESTAURANTS"},inplace=True)
rating_type_df
## Now we got number of restaurants by rating type now visuvalize this talble using chart to find percentage.
```

```
Out[64]:
```

	RATING_TYPE	Count_OF_RESTAURANTS
0	Average	4983
1	Good	4263
2	Very Good	1145
3	Excellent	96
4	Poor	56
5	Very Poor	4

```
In [65]: ## Add Relative frequency column too to find percentage of Restruarants.
# Total number of ratings
total_ratings = rating_type_df['Count_OF_RESTAURANTS'].sum()

# Add Relative Frequency column
rating_type_df['RELATIVE_FREQUENCY'] = rating_type_df['Count_OF_RESTAURANTS'] / total_ratings

# Display updated DataFrame
rating_type_df
```

```
Out[65]:
```

	RATING_TYPE	Count_OF_RESTAURANTS	RELATIVE_FREQUENCY
0	Average	4983	0.472457
1	Good	4263	0.404191
2	Very Good	1145	0.108562
3	Excellent	96	0.009102
4	Poor	56	0.005310
5	Very Poor	4	0.000379

```
In [66]: import plotly.express as px
fig = px.pie( rating_type_df, names='RATING_TYPE', values='RELATIVE_FREQUENCY', color='RATING_TYPE',
              title='Percentage of Restaurants by Rating Type' ).update_traces(textposition='inside', textinfo='
fig.show()
```

Q3, Which are the top 10 heighest rated seafood restaurant in mumbai?

```
In [67]: seafood_df= zomato_df[zomato_df['CUSINE_CATEGORY'].str.contains('Seafood')] #filter cusine_category where seafo
seafood_df.sort_values(by="RATING",ascending=False).head(10) #sort top 10 heighest rating seafood shop.
```

Out[67]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE_TYPE	TIMING	RATING_TYPE	RATING	VOTES	DA
7104	Thangabali	1000	Seafood,South Indian,Mangalorean,Andhra,Kerala	Khar	Bar	12noon to 4pm,7pm to 3am	Excellent	4.7	564	
76	Ceremonial Kitchen & Co	1000	Seafood,Maharashtrian,North Indian,Chinese	Thane	Casual Dining	1130am to 1130pm	Excellent	4.6	350	
13685	Maharashtra Lunch Home	600	Maharashtrian,Malwani,Konkan,Seafood	Kharghar	Casual Dining	11am to 345pm,7pm to 1145pm	Excellent	4.6	209	
12433	Quarter Canteen	1100	North Indian,Seafood,Chinese	Bandra	Casual Dining	12noon to 330pm,7pm to 1am	Excellent	4.5	573	
902	The Harbour Bay - SeaFood Kitchen & Bar	2400	Seafood,Beverages	Bandra	Casual Dining	12noon to 1am	Excellent	4.5	100	
884	Rajmanya-Seafood family restaurant	800	Maharashtrian,Konkan,Seafood	Vashi	Casual Dining	11am to 11pm	Excellent	4.5	178	
3380	Peco Peco	700	Chinese,Seafood,Asian	Powai	none	12noon to 330pm,7pm to 1230AM	Excellent	4.5	497	
9954	Pi Bar and Kitchen	1600	Continental,European,Italian,Seafood,Pizza,Des...	Andheri	Bar	12noon to 6pm,7pm to 12midnight	Excellent	4.5	2068	
903	Ferry Wharf	1500	Seafood,Mangalorean	Bandra	Casual Dining	11am to 330pm,7pm to 1230AM	Very Good	4.4	459	
915	Monis Bar and Restaurant	1000	Indian,Chinese,Continental,Seafood,Bever...	Thane	Casual Dining	1130am to 330pm,6pm to 1130pm	Very Good	4.4	662	

In [68]: `## There is no need to perform visuvalization for above quston because that's only differ in ratings.`

In [69]: `## MY TRY ##
Note only seafood restaurants are not good in rating.
seafood_df= zomato_df[zomato_df['CUSINE_CATEGORY']=='Seafood']
seafood_df.sort_values(by="RATING",ascending=False).head(10) #sort top 10 highest rating seafood shop.`

Out[69]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE_TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPENS
3826	Krishna Resto-Bar	600	Seafood	Thane	Casual Dining	12noon to 12midnight	Good	3.8	107	Mon-Sun
2080	Mukkam Post Konkan	600	Seafood	Navi Mumbai	Casual Dining	11am to 330pm,7pm to 11pm	Good	3.6	35	Mon-Sun
3527	Aieroc	800	Seafood	Alibaug	Casual Dining	12noon to 1130pm	Average	3.4	77	Mon-Sun
8602	Coastal Catch	1400	Seafood	Bandra	none	10am to 8pm	Average	3.3	7	Mon-Sun
12746	Aaditya Fish	300	Seafood	Kharghar	Quick Bites	1130am to 330pm,7pm to 1030pm	Average	3.3	18	Mon-Sun
2491	Saiva Gomantak	500	Seafood	Matunga	none	11am to 4pm,7pm to 12midnight	Average	3.2	52	Mon-Sun
12539	Sonal's Kitchen	400	Seafood	Malad	none	8am to 12noon,7pm to 12midnight	Average	3.1	7	Mon-Sun
7913	Guru Sawant's Malwan Katta	400	Seafood	Airoli	Quick Bites	11am to 4pm,6pm to 11pm	Average	3.0	4	Mon-Sun
7968	Karwar Fish Curry	700	Seafood	Thane	Casual Dining	12noon to 12midnight	Average	3.0	4	Mon-Sun
3987	Coastal Cafe	500	Seafood	Thane	Quick Bites	11am to 1230AM	Average	2.9	7	Mon-Sun

Q4, Which is the best food truck in mumbai?

In [70]: `foodtruck_df= zomato_df[zomato_df['CUSINE_TYPE']=='Food Truck'] #filter cuisine_type where foodtruck is only str
foodtruck_df.sort_values(by="RATING",ascending=False).head(2) #sort top 2 highest rating seafood shop.`

Out[70]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPENS
262	Dumpling Delights	200	Momos	Matunga	Food Truck	430pm to 930pm	Very Good	4.3	212	Mon-Sun
1017	Street Food Co.	250	Fast Food,Chinese	Virar	Food Truck	6pm to 3am	Very Good	4.1	274	Mon-Sun

Q5, Which places have the highest rated restaurent for each cuisine type in mumbai?

In [71]:

```
# Assuming restaurents having rating above 4.5
highestRatedDf=zomatoDf[zomatoDf["RATING"]>=4.5]
highestRatedDf
```

Out[71]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPEI
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	Bandra	Casual Dining	12noon to 130am	Excellent	4.9	3529	Mon-S
6	Persian Darbar	1300	Biryani,North Indian,Chinese,Mughlai	Marol	Casual Dining	10am to 3am	Excellent	4.5	3058	Mon-S
7	Tanatan	1500	Modern Indian	Juhu	Casual Dining	12noon to 130am	Excellent	4.7	1842	Mon-S
9	Plum by Bent Chair	1800	Asian	Lower Parel	Casual Dining	12noon to 1am	Excellent	4.7	1876	Mon-S
10	Angrezi Dhaba	1500	North Indian,Chinese,Thai,European	Dadar	Bar	12noon to 1am	Excellent	4.5	2092	Mon-S
...	...	...	...	...	...	...	...	...	...	...
14228	Zaika Crave - Club Aquaria	1300	North Indian,Continental,Chinese,Desserts	Borivali	Casual Dining	11am to 330pm,7pm to 1130pm	Excellent	4.5	1302	Mon,Tue,Wei
14234	Cone Culture	250	European	Kharghar	Casual Dining	Closed	Excellent	4.6	492	Mon,12noon 11pm(Tue-S
15007	Dessertino	300	Desserts,Ice Cream	Kandivali	Dessert Parlor	11am to 12midnight	Excellent	4.8	184	Mon-S
15051	Tick-eat	800	North Indian,Italian,Chinese,Mexican,Lebanese	Mulund	Casual Dining	1130am to 330pm,7pm to 1130pm	Excellent	4.5	754	Mon-S
15056	Daftar Goregaon	750	Pizza,Chinese,North Indian,Beverages	Goregaon	Casual Dining	12noon to 1130pm	Excellent	4.6	388	Mon-S

97 rows × 10 columns

In [72]:

```
fig = px.histogram(highestRatedDf, x='REGION', color='CUSINE TYPE',
                    title= 'No. of Best Restaurant for each Cuisine Type by Places').update_xaxes(categoryorder='ascending')
fig.show()
```

In [73]: highest Rated Restaurant Data

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPEN
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Biryani...	Bandra	Casual Dining	12noon to 130am	Excellent	4.9	3529	Mon-S
6	Persian Darbar	1300	Biryani,North Indian,Chinese,Mughlai	Marol	Casual Dining	10am to 3am	Excellent	4.5	3058	Mon-S
7	Tanatan	1500	Modern Indian	Juhu	Casual Dining	12noon to 130am	Excellent	4.7	1842	Mon-S
9	Plum by Bent Chair	1800	Asian	Lower Parel	Casual Dining	12noon to 1am	Excellent	4.7	1876	Mon-S
10	Angrezi Dhaba	1500	North Indian,Chinese,Thai,European	Dadar	Bar	12noon to 1am	Excellent	4.5	2092	Mon-S
...	...	...	...	...	...	...	...	...	...	...
14228	Zaika Crave - Club Aquaria	1300	North Indian,Continental,Chinese,Desserts	Borivali	Casual Dining	11am to 330pm,7pm to 1130pm	Excellent	4.5	1302	Mon,Tue,Wed
14234	Cone Culture	250	European	Kharghar	Casual Dining	Closed	Excellent	4.6	492	Mon,12noon to 11pm(Tue-S
15007	Dessertino	300	Desserts,Ice Cream	Kandivali	Dessert Parlor	11am to 12midnight	Excellent	4.8	184	Mon-S
15051	Tick-eat	800	North Indian,Italian,Chinese,Mexican,Lebanese	Mulund	Casual Dining	1130am to 330pm,7pm to 1130pm	Excellent	4.5	754	Mon-S
15056	Daftar Goregaon	750	Pizza,Chinese,North Indian,Beverages	Goregaon	Casual Dining	12noon to 1130pm	Excellent	4.6	388	Mon-S

97 rows × 10 columns

In [74]: *## Q6, What is the average price distribution of highest rated restaurant for each cuisine type in mumbai?*
highest Rated Restaurant Price Data
highest Rated Restaurant Price Data

	REGION	CUSINE TYPE	PRICE
0	Andheri	Casual Dining	1100.0
1	Andheri	none	350.0
2	Andheri	Bar	1600.0
3	Andheri	Café	800.0
4	Andheri	Casual Dining	1212.5

In [75]: fig = px.scatter(highest Rated Restaurant Price Data, x="REGION", y="PRICE", color="CUSINE TYPE", symbol="CUSINE TYPE",


```

title=' Avg Price Distobution of High rated restaurant for each Cuisine Type').update_traces(marker_
fig.show()

```

```

In [76]: zomato_df['CUISINE_CATEGORY'].unique()

```

```

Out[76]: array(['Modern Indian,North Indian,Chinese,Momos,Biryani,Continental,American,Fast Food',
        'Desserts,Ice Cream,Beverages', 'Asian,Chinese', ...,
        'Fast Food,Juices,Street Food',
        'Continental,Italian,Seafood,North Indian,Cafe',
        'Desserts,Chinese,Thai'], dtype=object)

```

```

In [77]: ## Q7,Which areas have a large number of chinese restaurent market?
chinese_df=zomato_df[zomato_df['CUISINE_CATEGORY'].str.contains('Chinese')]
#chinese_df = zomato_df[zomato_df['CUISINE_CATEGORY'].str.contains('Chinese')]
chinese_df.head()

```

```

Out[77]:

```

	NAME	PRICE	CUISINE_CATEGORY	REGION	CUISINE TYPE	TIMING	RATING_TYPE	RATING	VOTES	DAYS_OPENS
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	Bandra	Casual Dining	12noon to 130am	Excellent	4.9	3529	Mon-Sun
2	Chin Chin Chu	1800	Asian,Chinese	Juhu	Casual Dining	12noon to 1am	Very Good	4.2	337	Mon-Sun
4	BKC DIVE	1200	North Indian,Chinese,Continental	Bandra	Bar	1130am to 1am	Very Good	4.4	5995	Mon-Sun
5	Flea Bazaar Café	800	American,Asian,Street Food,North Indian,Luckno...	Lower Parel	Café	12noon to 1am	Very Good	4.2	2042	Mon-Sun
6	Persian Darbar	1300	Biryani,North Indian,Chinese,Mughlai	Marol	Casual Dining	10am to 3am	Excellent	4.5	3058	Mon-Sun

```

In [78]: ## Applies aggregation functions:
        ## 'NAME': 'count' → Counts how many restaurants are in each region.
        ## 'PRICE': 'mean' → Calculates the average price of Chinese restaurants in that region.
chinese_reset_df = chinese_df.groupby(by='REGION').agg({'NAME' : 'count', 'PRICE' : 'mean'}).rename(columns= {'
chinese_reset_df = chinese_reset_df.sort_values('Count of Restaurants', ascending=False).head(25)
chinese_reset_df.head()

```

```

Out[78]:

```

	REGION	Count of Restaurants	PRICE
79	Thane	348	588.376437
56	Mira Road	224	553.348214
50	Malad	186	604.032258
31	Goregaon	162	545.987654
4	Andheri	161	657.142857

```

In [79]: fig = px.bar(chinese_reset_df, x='REGION', y='Count of Restaurants', color='PRICE', title= 'No. of Chinese Rest

```

```
fig.show()
```

```
In [80]: ## Q8, Is there any relation between price and Rating by each cuisine type?
##Now we are going to create separate dataframe.
price_rating_df=zomato_df.groupby(['Cuisine Type','Rating'])['Price'].mean().reset_index()
price_rating_df ## For each group (e.g., "Italian, Rating=4"), it calculates the mean price.
                ##Produces a summary table where each row corresponds to a unique (Cuisine Type, Rating)
                ##pair with its average price.
```

```
Out[80]:
```

	Cuisine Type	Rating	Price
0	Bakery	2.7	400.000000
1	Bakery	2.8	285.714286
2	Bakery	2.9	328.571429
3	Bakery	3.0	300.000000
4	Bakery	3.1	369.117647
...	...	...	...
278	none	4.3	683.333333
279	none	4.4	555.000000
280	none	4.5	420.000000
281	none	4.6	687.500000
282	none	4.7	350.000000

283 rows × 3 columns

```
In [81]: ## visualization
fig=px.line(price_rating_df,y="Price",x='Rating',color='Cuisine Type') ##(line chart)
fig.show()
```

```
In [84]: ## Q9 is there a relation between Region and price?
region_price_df=zomato_df.groupby(["REGION"])[ "PRICE"].mean().reset_index()

## what is groupby?
## Each unique (Cuisine, Rating) becomes one group. here (region) only.
## Cuisine      Rating      Prices      Mean
##South Indian  4.0          200,180      190
```

```
In [85]: region_price_df
```

```
Out[85]:
```

	REGION	PRICE
0	Airoli	452.287582
1	Alibaug	665.000000
2	Ambarnath	514.000000
3	Andheri	579.779412
4	Andheri	592.993492
...	...	...
83	Vikhroli	480.434783
84	Vile Parle	464.457831
85	Virar	452.027027
86	Wadala	427.500000
87	Worli	537.012987

88 rows × 2 columns

```
In [86]: fig=px.scatter(region_price_df,x="REGION",y="PRICE").update_traces(marker_size=8) ## scatter plot
fig.show()
```

```
In [89]: ## Q10, Find the list of affordabe restaurants?
##The criteria of most affordabel restaurants would be:
#1. low price
#2. high rating
#Find low price (1/4 of high priced restaurents)
##example: most expensive restarent cost is 6k so one fouth of expencive is consider as an affordable price.
max_price=zomato_df["PRICE"].max()
aff_price=max_price/4
aff_price
```

Out[89]: 1250.0

```
In [91]: ## so affordable price is 1250
## extract subset of columns which is most important for our qustion.
aff_reset_df=zomato_df[["NAME","PRICE","CUSINE_CATEGORY","REGION","CUSINE TYPE"]]
aff_reset_df=aff_reset_df[aff_reset_df["PRICE"]<1250] ## filter by price
aff_reset_df.sort_values(by='PRICE',inplace=True) ## sort by price values.
aff_reset_df
```

Out[91]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE
6137	Sanjog Wine N Dine	5	North Indian,Chinese	Thane	Casual Dining
2925	Jab We Eat	50	South Indian,North Indian,Maharashtrian,Fast Food	Girgaum	none
9598	Ho5 Store	50	Fast Food	Matunga	none
9669	Himson Dryfruit & Sweets	100	Fast Food,Mithai	Vile Parle	Quick Bites
3127	Tandoor Chai	100	Fast Food,Beverages,Tea	Kalyan	Quick Bites
...	...	...	...	...	...
1837	Chi Na Chi Ni	1200	Asian	Kharghar	Casual Dining
14215	Reise All Day Bar & Kitchen	1200	Burger,American,Italian,North Indian,European,...	Chakala	Casual Dining
14214	Wild Dining Restaurant	1200	North Indian,Continental,Mexican,Chinese	Andheri	Casual Dining
7105	Bayview Cafe	1200	North Indian,American,Chinese	Colaba	Casual Dining
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	Bandra	Casual Dining

10187 rows × 5 columns

```
In [93]: ## 2, find high rated restaurants.
highrated_reset_df=zomato_df[["NAME","PRICE","CUSINE_CATEGORY","REGION","CUSINE TYPE","RATING"]] ## column subs
highrated_reset_df=highrated_reset_df[highrated_reset_df["RATING"]>4.5]
highrated_reset_df.sort_values(by="PRICE",inplace=True)
highrated_reset_df
```

Out[93]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	RATING
1502	Cake Centre-The Dessert Maker	150	Desserts	Mohammad Ali Road	Dessert Parlor	4.6

14234	Cone Culture	250	European	Kharghar	Casual Dining	4.6
149	The Northern Vibe	300	Momos,Rolls,Fast Food	Powai	Quick Bites	4.8
3924	Smiley Pops	300	Desserts,Ice Cream,Beverages,Sandwich	Andheri	Dessert Parlor	4.6
15007	Dessertino	300	Desserts,Ice Cream	Kandivali	Dessert Parlor	4.8
8893	Coppetto Artisan Gelato	350	Ice Cream,Desserts	Bandra	Dessert Parlor	4.6
901	Big Bang Cuurry	350	North Indian,Biryani,Rolls	Andheri	none	4.7
1796	TBG- The Biryani Guys	450	North Indian,Biryani	Powai	none	4.6
626	Hotel Sagar	500	North Indian,Chinese,Beverages	Chakala	Casual Dining	4.7
1177	Jazz & Blues	500	Cafe,Pizza,Italian	Andheri	Café	4.6
3757	B For Biryani	500	Biryani,North Indian,Kebab	Powai	Quick Bites	4.6
125	Too Much Drama	600	Fast Food,Roast Chicken,BBQ	Vashi	Casual Dining	4.7
59	Little West Pizza	600	Pizza	Borivali	Quick Bites	4.6
13685	Maharashtra Lunch Home	600	Maharashtrian,Malwani,Konkan,Seafood	Kharghar	Casual Dining	4.6
2573	Sam's Bohri Zaika	600	Bohri,North Indian,Mughlai,Kebab	Chandivali	Quick Bites	4.6
1847	Regano's	600	Continental,Fast Food,Italian,Desserts	Malad	Casual Dining	4.6
757	Cookstory	700	North Indian,Chinese,Mughlai	Andheri	none	4.6
202	Dum & Curry	700	Mughlai,North Indian,Chinese	Powai	Quick Bites	4.7
1533	Fruitilicious	700	Desserts	Kalbadevi	Dessert Parlor	4.6
15056	Daftar Goregaon	750	Pizza,Chinese,North Indian,Beverages	Goregaon	Casual Dining	4.6
66	Downtown China	750	Chinese,Thai	Andheri	Casual Dining	4.8
3026	Makhan Singh	800	North Indian,Chinese,Biryani	Powai	none	4.6
1251	Joey's Pizza	800	Pizza	Malad	Quick Bites	4.6
588	Pepper Fry	800	Chinese,Italian,South Indian,North Indian	Thane	Casual Dining	4.6
1968	Curry Culture	800	North Indian,Biryani,Chinese,Kebab,Mughlai,Asian	Powai	none	4.6
13859	Family Tree	800	Italian,Mexican,North Indian,Chinese,Salad	Thane	Casual Dining	4.6
9778	Rajdhani	950	Gujarati,Rajasthani,North Indian	Ghatkopar	Casual Dining	4.8
4099	Butter Chickenwala	1000	North Indian,Chinese,Biryani	Powai	Quick Bites	4.7
76	Ceremonial Kitchen & Co	1000	Seafood,Maharashtrian,North Indian,Chinese	Thane	Casual Dining	4.6
12094	Sandy's Den	1000	Fast Food,Bar Food	Chembur	Casual Dining	4.6
7104	Thangabali	1000	Seafood,South Indian,Mangalorean,Andhra,Kerala	Khar	Bar	4.7
724	Maezo	1000	Modern Indian	Thane	Casual Dining	4.6
1080	The Poshpit	1100	Cafe,Continental,Chinese,Mexican,Pizza,Salad,B...	Andheri	Café	4.6
369	Aquafire Restaurant	1100	North Indian,Continental,Chinese,Italian	Vile Parle	Casual Dining	4.6
0	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	Bandra	Casual Dining	4.9
196	Culinary Tales	1200	Chinese,European,Continental,Salad,Italian,Pizza	Andheri	Casual Dining	4.7
50	Spice Republic	1200	Cafe,Continental,Mediterranean,Mexican,Italian...	Borivali	Café	4.6
2321	Lion Heart	1400	North Indian,Chinese,Italian,Finger Food,Asian...	Byculla	Lounge	4.6
5156	Paps Premium Lounge	1400	North Indian,Mexican,Continental	Thane	Lounge	4.7
459	Royale Masterchef Finedine and Bar	1400	North Indian,Asian,Continental,Lebanese,Salad	Dahisar	Casual Dining	4.6
12	Todi Mill Social	1400	American,North Indian,Chinese,Fast Food,Contin...	Lower Parel	Bar	4.6
3554	Smoke House Deli	1400	European,Italian,Salad,American,Burger,Juices,...	Bandra	Casual Dining	4.7
12440	The Stables - Peninsula Redpine	1500	American,Italian,Mexican,Continental	Marol	Bar	4.6
12636	Khow Chow	1500	Asian	Bandra	Casual Dining	4.6
7	Tanatan	1500	Modern Indian	Juhu	Casual Dining	4.7
824	Blend N Brew	1600	North Indian,Chinese,American	Navi Mumbai	Casual Dining	4.7
8185	The Fusion Kitchen	1600	Chinese,Continental,European,North Indian,Ital...	Marol	Casual Dining	4.7
10669	Trumpet Sky Lounge	1700	North Indian,Chinese	Andheri	Lounge	4.9
1781	Yazu - Pan Asian Supper Club	1700	Thai,Korean,Japanese,Chinese	Andheri	Casual Dining	4.6
9	Plum by Bent Chair	1800	Asian	Lower Parel	Casual Dining	4.7
1786	Global Fusion	2000	Chinese,Japanese,Asian,North Indian	Worli	Fine Dining	4.6

In [98]: `## Now we have aff price dataframe and highrated dataframe to create affordable dataframe we should merge that
highrated_aff_df=pd.merge(aff_reset_df,highrated_reset_df,how='inner',on=['NAME','REGION'])
##Join type: how='inner' → keeps only rows where both DataFrames have matching values.`

```
##Join keys: on=['NAME', 'REGION'] → the merge is performed using restaurant name and region as the common iden
```

```
In [99]: highrated_aff_df = highrated_aff_df[['NAME', 'PRICE_x', 'CUSINE_CATEGORY_x', 'REGION', 'CUSINE TYPE_x']]
## When two DataFrames have overlapping column names, Pandas automatically appends suffixes (_x and _y)
## to distinguish them.
```

```
In [100]: highrated_aff_df.rename(columns={'NAME':'NAME', 'PRICE_x':'PRICE', 'CUSINE_CATEGORY_x':'CUSINE_CATEGORY',
                                         'REGION':'REGION', 'CUSINE TYPE_x':'CUSINE TYPE'},inplace=True)
## This step removes unnecessary or duplicate columns
## (example 'PRICE_x','CUSINE_CATEGORY_x','CUSINE TYPE_x') after the merge.
```

```
In [101]: highrated_aff_df
```

Out[101]:	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE
0	Cake Centre-The Dessert Maker	150	Desserts	Mohammad Ali Road	Dessert Parlor
1	Cone Culture	250	European	Kharghar	Casual Dining
2	Smiley Pops	300	Desserts,Ice Cream,Beverages,Sandwich	Andheri	Dessert Parlor
3	The Northern Vibe	300	Momos,Rolls,Fast Food	Powai	Quick Bites
4	Dessertino	300	Desserts,Ice Cream	Kandivali	Dessert Parlor
5	Big Bang Cuurry	350	North Indian,Biryani,Rolls	Andheri	none
6	Coppetto Artisan Gelato	350	Ice Cream,Desserts	Bandra	Dessert Parlor
7	TBG- The Biryani Guys	450	North Indian,Biryani	Powai	none
8	Hotel Sagar	500	North Indian,Chinese,Beverages	Chakala	Casual Dining
9	B For Biryani	500	Biryani,North Indian,Kebab	Powai	Quick Bites
10	Jazz & Blues	500	Cafe,Pizza,Italian	Andheri	Café
11	Too Much Drama	600	Fast Food,Roast Chicken,BBQ	Vashi	Casual Dining
12	Little West Pizza	600	Pizza	Borivali	Quick Bites
13	Regano's	600	Continental,Fast Food,Italian,Desserts	Malad	Casual Dining
14	Maharashtra Lunch Home	600	Maharashtrian,Malwani,Konkan,Seafood	Kharghar	Casual Dining
15	Sam's Bohri Zaika	600	Bohri,North Indian,Mughlai,Kebab	Chandivali	Quick Bites
16	Dum & Curry	700	Mughlai,North Indian,Chinese	Powai	Quick Bites
17	Cookstory	700	North Indian,Chinese,Mughlai	Andheri	none
18	Fruitionicious	700	Desserts	Kalbadevi	Dessert Parlor
19	Downtown China	750	Chinese,Thai	Andheri	Casual Dining
20	Daftar Goregaon	750	Pizza,Chinese,North Indian,Beverages	Goregaon	Casual Dining
21	Pepper Fry	800	Chinese,Italian,South Indian,North Indian	Thane	Casual Dining
22	Family Tree	800	Italian,Mexican,North Indian,Chinese,Salad	Thane	Casual Dining
23	Makhan Singh	800	North Indian,Chinese,Biryani	Powai	none
24	Curry Culture	800	North Indian,Biryani,Chinese,Kebab,Mughlai,Asian	Powai	none
25	Joey's Pizza	800	Pizza	Malad	Quick Bites
26	Rajdhani	950	Gujarati,Rajasthani,North Indian	Ghatkopar	Casual Dining
27	Maezo	1000	Modern Indian	Thane	Casual Dining
28	Thangabali	1000	Seafood,South Indian,Mangalorean,Andhra,Kerala	Khar	Bar
29	Sandy's Den	1000	Fast Food,Bar Food	Chembur	Casual Dining
30	Ceremonial Kitchen & Co	1000	Seafood,Maharashtrian,North Indian,Chinese	Thane	Casual Dining
31	Butter Chickenwala	1000	North Indian,Chinese,Biryani	Powai	Quick Bites
32	Aquafire Restaurant	1100	North Indian,Continental,Chinese,Italian	Vile Parle	Casual Dining
33	The Poshpit	1100	Cafe,Continental,Chinese,Mexican,Pizza,Salad,B...	Andheri	Café
34	Spice Republic	1200	Cafe,Continental,Mediterranean,Mexican,Italian...	Borivali	Café
35	Culinary Tales	1200	Chinese,European,Continental,Salad,Italian,Pizza	Andheri	Casual Dining
36	Hitchki	1200	Modern Indian,North Indian,Chinese,Momos,Birya...	Bandra	Casual Dining

Q10) Find the list of most Reliable Restaurants?

The criteria for most Reliable Restaurants would be:

##1) Low Price 2) High Rated 3) Large No. of Votes

First step will be to find the restaurants with Votes greater than Mean of Votes

```
In [102]: mean_votes = zomato_df['VOTES'].mean()  
mean_votes
```

Out[102]: 177.2656679624538

```
In [103]: # Finding list of restaurants that have Votes greater than and equal to Mean of Vote  
  
mean_rest_df = zomato_df[['NAME', 'PRICE', 'CUSINE_CATEGORY', 'REGION', 'CUSINE TYPE', 'VOTES']]  
mean_rest_df = mean_rest_df[mean_rest_df['VOTES'] > 177]  
mean_rest_df.sort_values(by='VOTES', inplace=True)  
mean_rest_df
```

Out[103]:

	NAME	PRICE	CUSINE_CATEGORY	REGION	CUSINE TYPE	VOTES
4194	Sai Sagar Veg Treat	500	North Indian,South Indian,Chinese,Fast Food,Be...	Kalyan	Casual Dining	178
884	Rajmanya- Seafood family restaurant	800	Maharashtrian,Konkan,Seafood	Vashi	Casual Dining	178
3914	Ice Cafe	500	Fast Food,Ice Cream,Beverages,Pizza	Borivali	Quick Bites	178
7897	Konkan Lajjatdar	500	Seafood,Biryani,Beverages,Chinese,Malwani,Konkan	Andheri	Casual Dining	178
3828	Frozen Delight -The Dessert Cafe	250	Desserts,Ice Cream	Airoli	Dessert Parlor	178
...	...	...	...	...	...	...
8539	Leopold Cafe & Bar	1600	American,Chinese,Mughlai,Italian	Colaba	Casual Dining	7327
1251	Joey's Pizza	800	Pizza	Malad	Quick Bites	7350
5337	Chili's American Grill & Bar	1400	American,Mexican,Burger,Tex-Mex	Powai	Casual Dining	7377
3751	Prithvi Cafe	700	Cafe,Fast Food	Juhu	Café	8000
8897	Candies	700	Cafe,Italian,North Indian,Desserts	Bandra	Café	10217

2345 rows × 6 columns

```
In [ ]:
```